# Evaluation of the Use of INTERPRET Decision Support System Software V3.1 in the Diagnosis of Brain Lesions Based on Single Voxel Magnetic Resonance Spectroscopy


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## INTRODUCTION

Magnetic Resonance Spectroscopy (MRS) is frequently used as a complimentary diagnostic tool in the diagnostic workup of brain lesions. However, its interpretation is usually challenged by the experience of radiologists adding more time, and effort on them.

The **International Network for Pattern Recognition of Tumors Using Magnetic Resonance Decision Support System (INTERPRET DSS)*** is a software developed by a multi-center European collaboration. The INTERPRET DSS software incorporates a database of large number of long and short TE Single Voxel (SV) MRS of diagnosed brain lesions confirmed by histopathology.

## AIM OF STUDY

To evaluate the use of INTERPRET DSS V3.1 software in the diagnosis of brain lesions based on SV-MRS in terms of its diagnostic accuracy and the time needed to make the diagnosis.

## METHODS

- Following IRB approval, we included 25 patients (mean age= 40.3 yrs, SD=15.8) who presented with undiagnosed intracranial space occupying lesions. Each patient underwent a brain SV-MRS study on 1.5T MR scanner. Our MRS PRESS protocol included both long TE (144 ms) and short TE (30 ms) sequences.
- MR spectra were extracted in a DICOM format and then further processed and converted as required for analysis by INTERPRET DSS software.
- Two groups of radiologists reviewed the MRS studies; the first group consisted of two certified radiologists with comparable experience independently reviewed the cases "**unaided**" by the software. Afterwards another group formed by a third certified radiologist (with a similar experience ) blindly reviewed the same cases but "**aided**" by the software.

## INTERPRET DSS V3.1 software

The software has three classification interfaces based on short TE, long TE, or both:

A- Distinction among most common tumor types: The classifiers were trained with the following brain tumor superclasses: low grade meningiomas, low grade glial tumors (low grade astrocytoma and oligodendroglioma) and high-grade tumors (anaplastic astrocytoma or oligodendroglioma, GBM, and Metastasis).

B- Distinction between tumor and pseudotumoral disease: The Classifiers differentiate between pseudotumoral disease, tumors and normal brain tissue.

C- Distinction between GBM and metastases: This method differentiates GBM and metastatic lesions.

Case 1 is shown below as an example (figure 1).

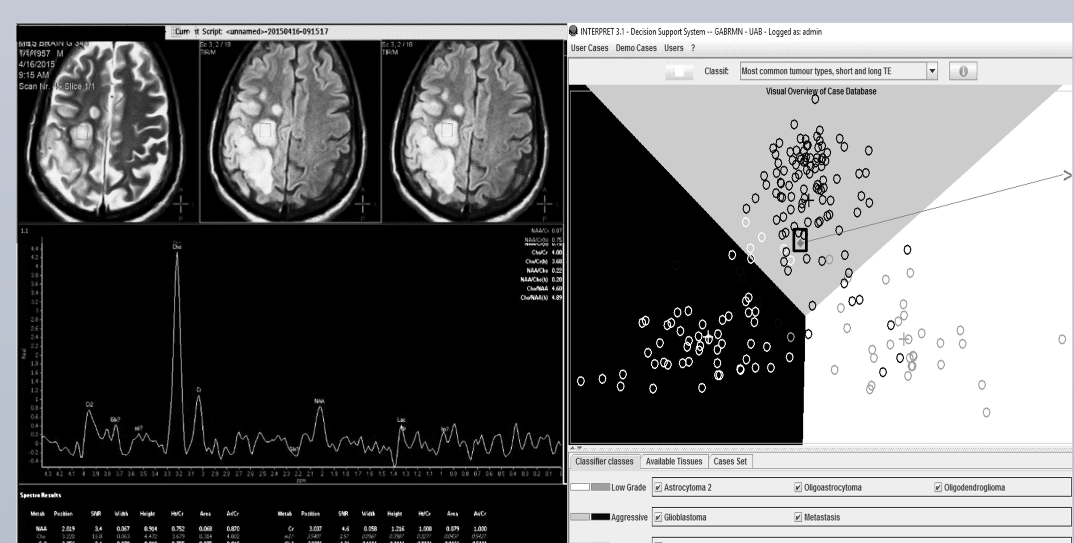

**Figure 1** (Case 1): A 58 year-old male with right frontoparietal GBM seen on T2 and FLAIR axial images (left upper) and SV-MRS (left lower) using long TE (144 ms). INTERPRET DSS v3.1 interface (right) classified the MRS pattern as an aggressive (high grade) neoplasm (small rectangle near the center of the image).

## RESULTS

Among the 25 included patients, a diagnosis of 16 neoplastic and 9 non-neoplastic lesions was confirmed by pathology or clinical consensus (when pathology was not indicated) (table 1). ROC analysis showed better performance of group 3 (RAD3) aided by INTERPRET DSS (AUC = 1.0) in differentiating neoplastic versus non neoplastic lesions compared to the unaided radiologist 1 (RAD1) or 2 (RAD2), p = 0.01 (figure 2-left). This better performance of group 3 (RAD3) aided by the software was even stronger in cases with low grade neoplastic lesions (figure 2-right).

Radiologists 1 and 2 (unaided by the software) had an inter-rater agreement of 83% (kappa = 0.75, p = 0.0001).

The time required to make a diagnostic impression based on SV-MRS was relatively shorter in group 3 aided by the software compared to the unaided group (figure 3).

| Diagnosis | N |
|---|---|
| Low grade glioma | |
| Juvenile pilocytic astrocytoma | 1 |
| Low grade glioma Grade II | 4 |
| High grade neoplasm | |
| GBM | 4 |
| Anaplastic astrocytoma Grade III | 3 |
| Hemangiopericytoma | 1 |
| Metastasis | 3 |
| Non neoplastic lesions | |
| ADEM | 1 |
| Infarction | 2 |
| Vasculitis | 2 |
| Radiation necrosis | 4 |
| Total | 25 |

**Table 1:** Frequency of cases according to final diagnosis.

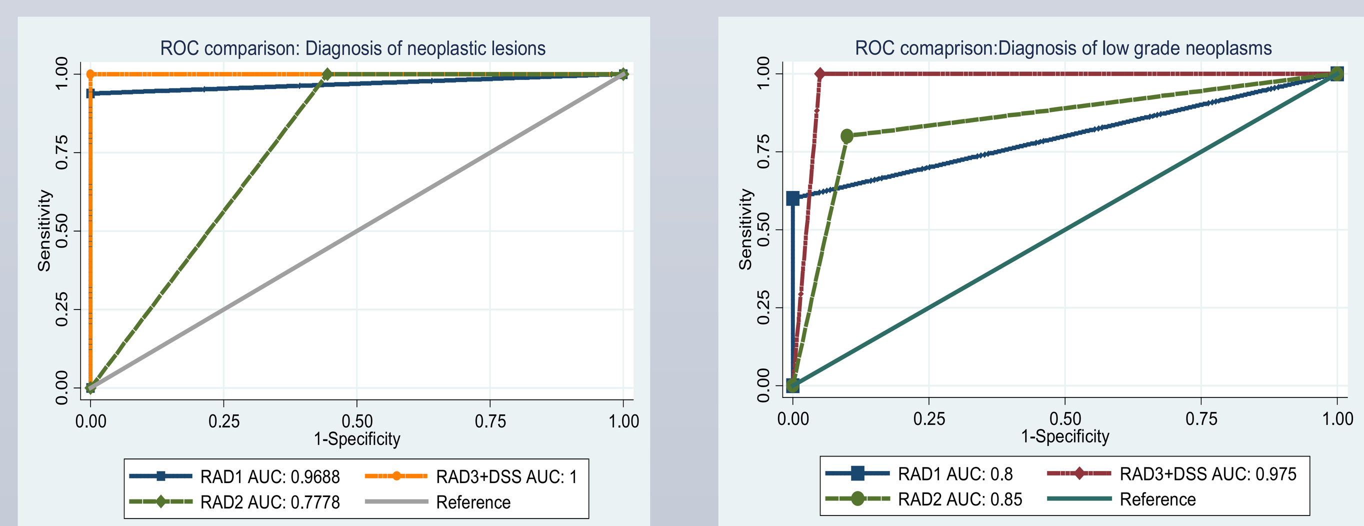


**Figure 2:** Comparison of ROC analyses showing better performance of Radiologist 3 aided by INTERPRET DSS software (RAD3+DSS) versus the two unaided radiologists RAD1 or RAD2 in making a true diagnosis in overall neoplastic lesions (left) or low-grade neoplastic lesions (right). AUC: Area under the curve.

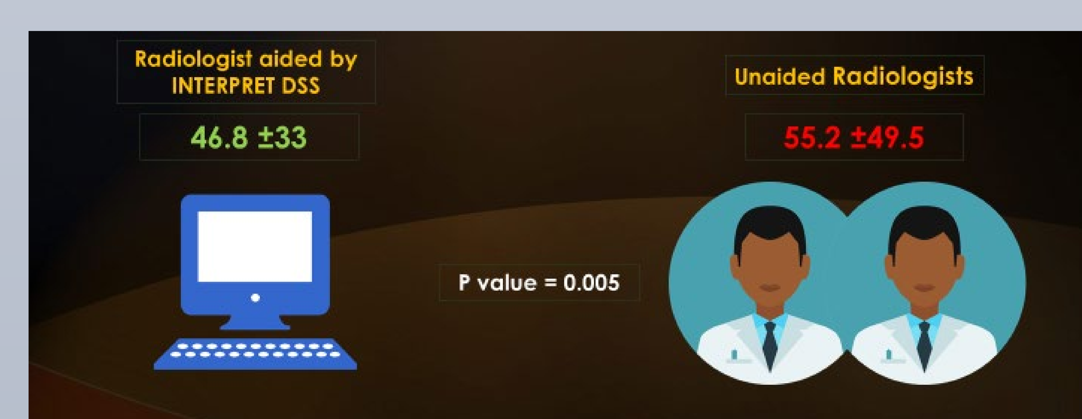


**Figure 3:** Comparison of the mean time taken by the radiologist aided by software (46.8 sec) versus the two unaided radiologists (55.2 sec) to make a diagnostic impression by reviewing SV-MRS.

## OTHER ILLUSTRATED CASES

**Case No.2:**

A 39-year old female presented with disturbed level of consciousness. There are multiple bilateral cerebral white matter un-enhancing T2 hyperintense lesions with Cho/NAA = 0.97 (figure 4-left). INTERPRET DSS software placed the lesion in the non-neoplastic group (figure 4-right). Final diagnosis was ADEM.

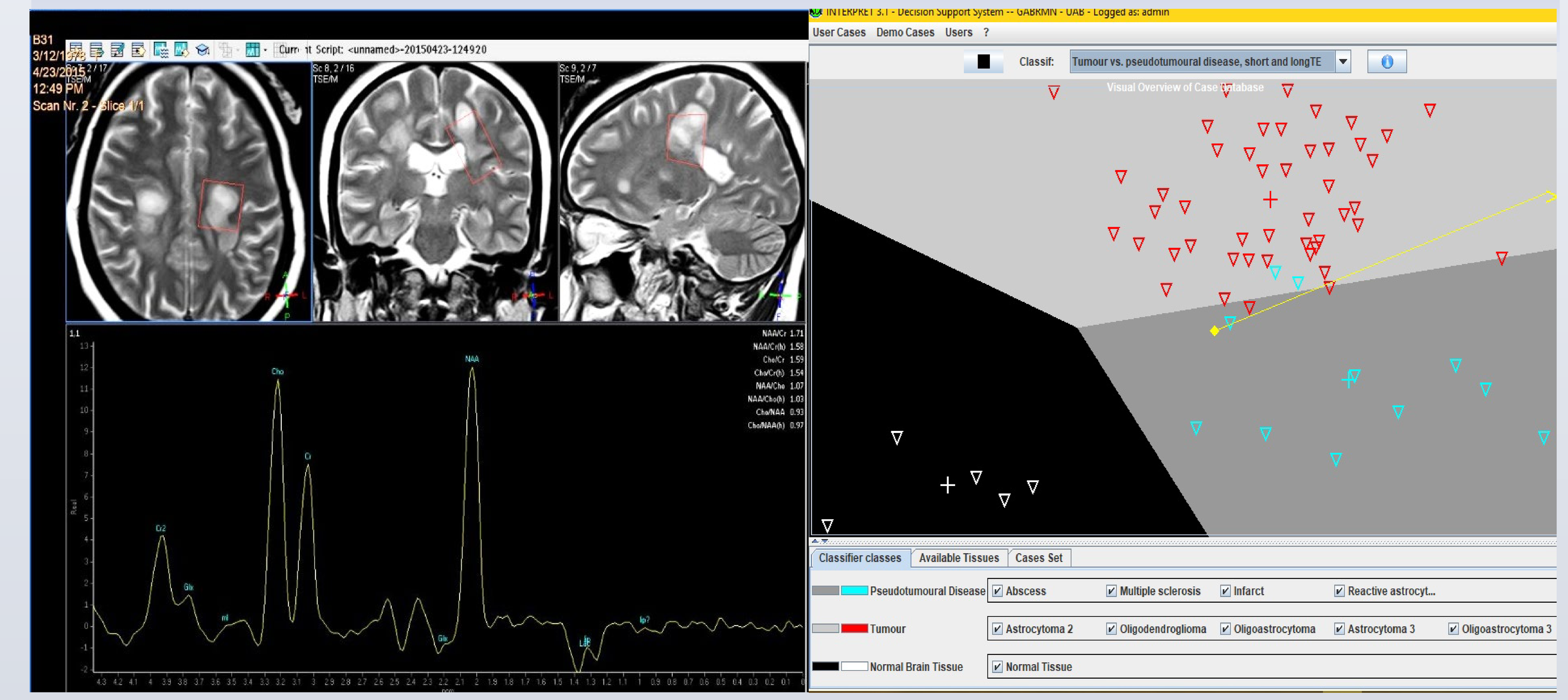

**Figure 4: T2, SV-MRS (left) and INTERPRET DSS classifier display (right) in a case of ADEM.**

**Case No.3:**

A 31-year old female presented with seizures. A bright T2 space occupying lesion is seen at right temporal lobe with Cho/NAA ratio = 2.33 (figure 5-left). INTERPRET DSS software placed the lesion in the low grade neoplastic group (figure 5-right). Final diagnosis was low grade glioma.

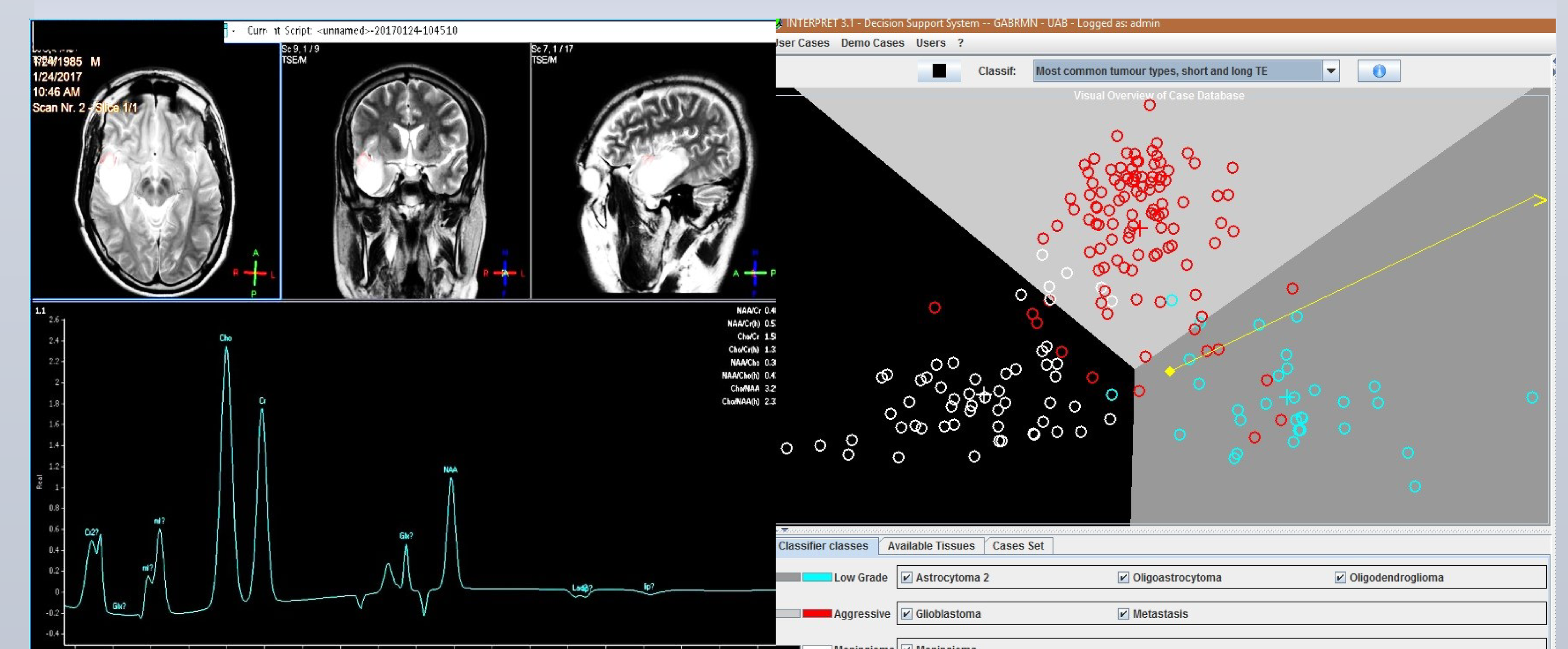

**Figure 5: T2, SV-MRS (left) and INTERPRET DSS classifier display (right) in a case of right temporal low grade glioma.**

## CONCLUSION

**The use of INTERPRET DSS V3.1 software in the diagnosis of brain lesions based on single voxel MR spectroscopy has relatively reduced the time needed for diagnosis and improved the diagnostic performance of radiologists.**


*Alexander Pérez-Ruiz, Margarida Julià-Sapé, Guillem Mercadal, Iván Ollier, Carles Majós and Carles Arús. BMC Bioinformatics 2010, 11:58. DOI: 10.1186/1471-2105-11-581